Photonic plug for scalable silicon photonics packaging

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ABSTRACT

Scalability of silicon photonics packaging is required to support the growing demand for bandwidth in data centers and other emerging datacom and telecom applications. Connecting fiber to chip is still considered one of the main challenges of silicon photonics today due to the need for high alignment accuracy, which in turn requires expensive assembly machines and in most cases active alignment protocols. Furthermore, current fiber assembly technologies are not suitable for multiple emerging applications with large port count such as co-packaged optics. We present here the Photonic-plug technology, a unique self-aligning optical arrangement, which enables large assembly tolerance suitable for passive alignment protocols and for scalable silicon photonics port count packaging. The Photonic-plug technology accomplishes die stacking geometry with efficient wideband surface coupling as well as with grating coupler based silicon photonic chips. The combination of the Photonic-plug's large assembly tolerances and surface coupling geometry enables efficient silicon photonics wafer level testing capabilities prior to dicing. The Photonic-plug technology takes advantage from wafer level fabrication processes for planar and accurate implementing of optical elements. A library service model, called Photonic-bump, is incorporated as part of the Photonic-plug technology through silicon photonic wafer manufacturing process for complete removal of the fibers' mechanical constraints from wafer manufacturing process. The Photonic-plug takes fiber-to-chip packaging away from specialized, low throughput and expensive tools to standard, automated and high volume flip-chip packaging machines. Standardizing optical packaging through Photonic-plug methodology will affect further silicon photonics application to thrive such as photonic FPGA, optical interposers and chip-to-chip optical connectivity.

Keywords: Silicon photonics, photonic packaging, fiber assembly, co-packaged optics

1. INTRODUCTION

Silicon photonics wafer manufacturing is aligned well with the already established microelectronics fabrication facilities and benefits its superior scalability of volume production, high yields and low costs [1-5]. However, fiber to chip packaging is still a big challenge for silicon photonics to thrive as a platform and to fulfill its huge potential in datacom, telecom, sensors and other promising applications due to lack of optical packaging scalability. Current single mode fiber (SMF) assembly methodologies use expensive and low throughput specialized equipment which are not compatible with standard and high volume chip packaging lines [6-8]. Mode field diameter (MFD) mismatch between SMF's core and silicon photonics waveguides as well as the unknown polarization of the former compared to the strong polarization confinement of the later are key factors that generate the alignment obstacle. Various industrial and research attempts have been studied to address this obstacle and can be divided into two main categories; a) surface coupling geometry and b) edge coupling geometry. Surface coupling geometries are mostly based on grating coupler elements [9-14] which exhibit relatively relaxed assembly tolerances, convenient planar lithographic fabrication, wafer level testing and do not involve additional post processing steps. These advantages are catalysts for the industrial adoption of surface coupling geometries such as in PSM4 100G transceiver devices [15]. On the other hand, grating couplers impose strong wavelength and polarization dependencies thus limit their adoption in silicon photonics broadband related applications. Edge coupling geometries involve variety of mode-converter methodologies incorporated on the silicon photonic wafers for adiabatic coupling to SMFs.

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These include inverse tapered silicon waveguide integrated with spot size converter (SSC) polymeric materials [16,17], silicon oxynitride [16], silicon-rich oxide [18] or highly efficient planar sub-wavelength metamaterial gratings [19,20]. Several edge coupling works have shown a low insertion loss and passive or semi-passive fiber alignment capabilities [21-24]. Edge coupling presents advantages over grating couplers in terms of compatibility with broad band, polarization independent and a better insertion loss performance [21,22]. However, it limits the scalability to obtain high volume production due to several complexities including precision dicing and polishing of the silicon photonics die, tight of alignment tolerances, mechanical constraints and incompatibility to perform high fiber count assembly [25]. In our view, while these techniques of efficiently inserting light into the silicon device layer either by grating coupler or by inverted silicon taper are indispensable, they are not sufficient to scale up silicon photonics platform to a full fabless path as they require major deviation from the silicon outsourced assembly and test (OSAT) standard manufacturing flow. In order to comply with OSAT's requirement the fiber connection should be similar to die stacking methodologies, that is; surface attachment with many microns of tolerance.

We present here the Photonic-plug technology, an add-on to the grating coupler and inverted taper techniques for light injection to silicon channels. In this technology, the fiber assembly tolerances are significantly improved by placement of optical elements in wafer level for both broadband and grating coupler based silicon photonic chip. The Photonic-plug operation is based on a "self-aligning optics" scheme [26] which takes advantage from planar and accurate wafer level implementation of optical elements. The optical scheme is presented in Fig. 1(a) and shows a confocal beam-tracing diagram that images the fiber's core to the photonic chip surface. The upper side of the diagram represents the Photonic-plug die which combines fiber end (blue block), tilted "1" and curved "2" mirrors that are perfectly aligned to each other. The bottom side of the scheme represents the silicon photonic chip with additional optical elements "3" (called Photonic-bump) implemented via accurate wafer level process next to the vertical coupling element "4" (grating coupler or a turning mirror when wide-band silicon photonic chip with inverse taper is used, see more details in section 2.2). A transparent medium with appropriate index of refraction and thickness values separates the Photonic-plug die and the silicon photonic chip. Such confocal scheme exhibits large spatial tolerances upon relative displacement between Photonic-plug and silicon photonic chip as presented in the Fig.1(b) which shows a simulation of added loss due to XYZ displacements for such geometry. The source for this expanded tolerance is the symmetry of the system, where an offset of the fiber relative to the curved mirror "3" is fully compensated by an identical offset of the Photonic-plug curved mirror "2" relative to the silicon photonic chip's coupler "4" (see optical scheme Fig. 1(a)).

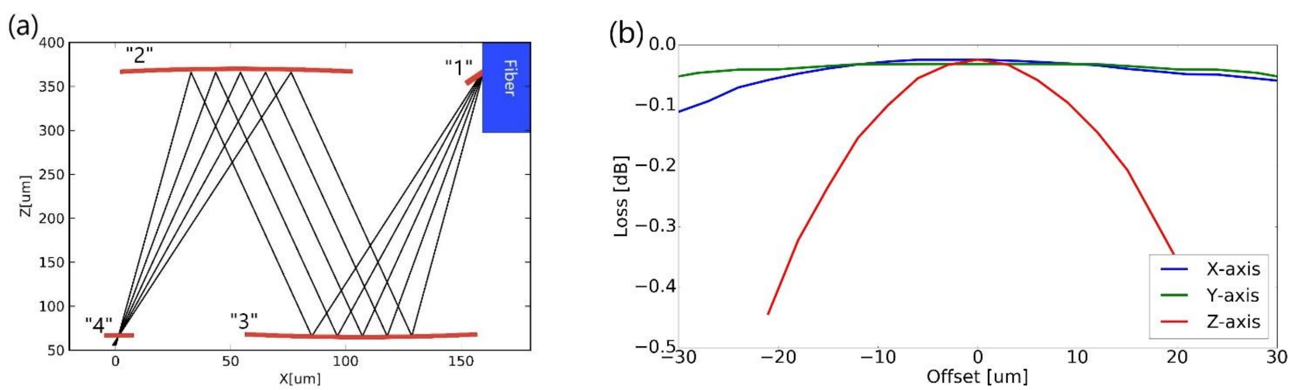

Figure 1: (a) Photonic-plug "self-aligning" optical scheme. Accurate placement of optical elements via wafer level processes combining Photonic-plug die with fiber (fiber is presented as blue block), flat tilted mirror "1" and a curved mirror "2" on the top side and a silicon photonic chip with Photonic-bump "3" and vertical coupling element "4" on the bottom side of the scheme. (b) Simulation of the added loss due to XYZ displacement of the Photonic-plug relative to the silicon photonic chip.

In summary, the Photonic-plug methodology moves the fiber assembly complexity and accuracy requirements to the wafer level manufacturing domain. In this domain, high accuracy can be easily achieved for both the fiber block wafer (Photonic-plug dies) and the silicon phonic wafer. After qualifying the fabrication tolerances at wafer level, the assembly tolerances are loosen to support fast and cost effective assembly machines by passive alignment protocols. To fulfill this production flow, the Photonic-bump is suggested as a library element of the silicon photonics wafer foundry provider or OSAT vendors similarly to standard electronics bumps model and thus aligns optical packaging with microelectronics fabrication and packaging flow. Photonic-plug technology removes edge coupling complexity and uses surface coupling geometry with large assembly tolerances which are suitable for passive assembly and high throughput flip-chip machines.

Furthermore, the combination of convenient surface coupling geometry and large tolerances marks the Photonic-plug as a great candidate for assembling large number of fibers and allow for pluggable fiber assembly solutions for applications such as co-packaged optics [27]. We claim that Photonic-plug has the potential to scale up silicon photonics packaging and align it with standard design, manufacturing and packaging industrial lines.

2. METHODOLOGY

2.1. Photonic-plug fabrication

The Photonic-plug is a fiber block device which composes v-groove trenches that are perfectly aligned to three-dimensional optical elements including flat and concave mirrors (Fig. 2(a)). Wafer level fabrication methodologies are used to produce the Photonic-plug die, to set the geometry of mirrors and to ensure accurate placement of the different elements. Two fabrication methodologies have been used: a) standard complementary metal oxide semiconductor (CMOS) manufacturing process that incorporates gray scale lithography (achieved by standard half-tone chromium mask), dry etching and wet etching processes and b) ultra-precision molding via wafer level optics replication process. Metal deposition (Au or Al) is then performed through evaporation process to achieve high optical reflectivity from mirrors' surface. Glass spacer (Borofloat 33) having 250um thickness is then attached to the Photonic-plug by die to wafer placement. Mechanical pedestals are produced as part of the Photonic-plug wafer fabrication process to ensure high leveling accuracy of glass attach process. Wafer dicing process is then performed to singulate the Photonic-plug dies for fiber assembly process.

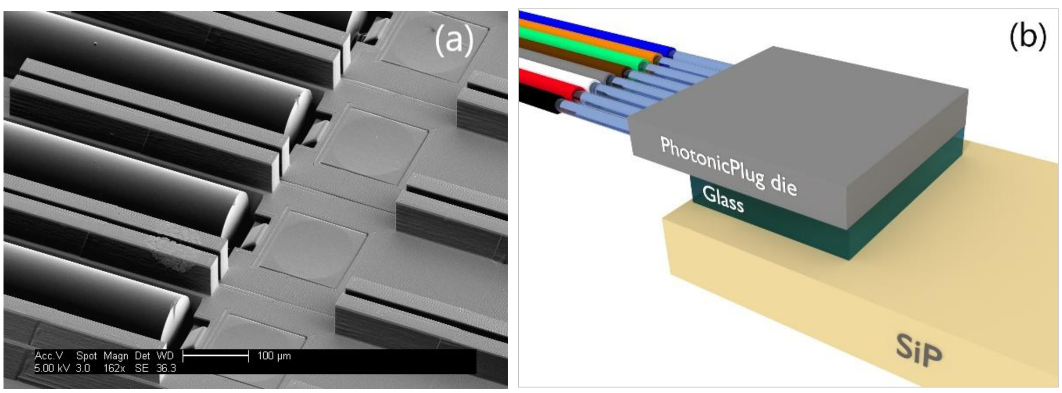

Figure 2: (a) Scanning electron microscopy (SEM) image of a Photonic-plug die showing a section of single mode fiber array assembled inside v-grooves in close proximity to flat and concave mirrors. Image bar is 100µm (b) A scheme of a Photonic-plug device including a Photonic-plug die (facing down), a single mode fiber array and a glass spacer placed on a silicon photonics chip.

Single mode fiber ribbon with typical mechanical cleavage is inserted into v-groove trenches by passive assembly process. Mechanical stoppers, which are produced as part of the wafer fabrication process, are placed at the end of the v-grooves to determine the end position of the fiber inside the trench. Fiber cleavage inaccuracies might generate a varying distance between fiber facet and tilted mirror of up to 10µm for an array with 12 fibers. The contribution of these inaccuracies to the loss is low as those inaccuracies are well below the 60µm Rayleigh distance of single mode fiber beam in matching index environment. UV cured adhesive (Noa 86) is used to secure the fibers inside the v-grooves and to ensure complete adhesive fill between fibers, mirrors and the glass spacers with appropriate index of refraction. A scheme of an assembled Photonic-plug including a Photonic-plug die, single mode fiber ribbon and a glass spacer placed on a silicon photonics chip is shown on Fig. 2(b).

2.2. Photonic-bump fabrication

The Photonic-bump consists optical elements which are implemented on the silicon photonics wafer and by wafer level processes to guarantee accurate placement of these elements relative to the silicon channel. For a grating coupler-based chip, a grating coupler Photonic-bump (GCPB) is used to perform the "self-aligning" optics scheme. The GCPB is implemented as part of the silicon photonics wafer manufacturing process and involves gray scale lithography by standard half-tone chromium mask, dry etching and metal coating processes. Fig. 3 shows an example of a GCPB, composed of curved mirror which is placed accurately in close proximity to the grating coupler element on a silicon photonic test chip of LETI-CEA Tech (Technology Research Institute, France). A gold coated GCPB elements, placed next to the input and output of grating couplers on the test chip are shown in Fig. 3(a). Fig. 3(b) shows three dimensional optical profiler measurement of such curved mirror. The distance between the GCPB and the grating coupler was determined according to the chief ray intersections of the "self-aligning" optics diagram which depends on various free parameters such as spacer thickness and grating coupler emission angle at center wavelength.

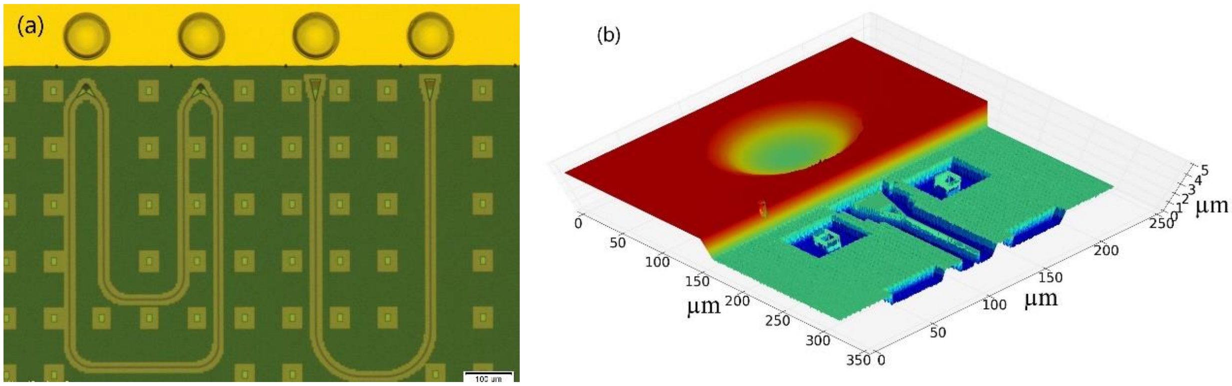

Figure 3: (a) Silicon photonics test chip of LETI-CEA with Photonic-bumps implemented via wafer level process next to grating coupler elements. Bar is 100µm (b) Optical profiler measurement of a typical Photonic-bump next to a grating coupler element.

In the case of wide-band chip, inverse taper Photonic-bump [ITPB] type is used to perform three tasks: a) To transfer fiber assembly process from edge coupling to surface coupling geometry, b) to perform spot size conversion (SSC) between SMF core and silicon channel and c) to enable large assembly tolerances through the Photonic-plug "self-aligning optics" setup. ITPB incorporates three main elements: a) positive tapered waveguide (labeled as "1" in Fig. 4(a)), b) turning mirror (labeled as "2" in Fig. 4(a)) for surface coupling geometry and c) curved mirror (labeled as "3" in Fig. 4(a)) for obtaining the "self-aligning optics" scheme. The ITPB waveguide structure begins with a cross section of 2µm height and 5µm width and ends after adiabatic SSC in three dimensions along 500µm length to match an SMF beam shape. A silicon photonics circuit with test features was manufactured by Advanced Micro Foundry (AMF), Singapore. The test chips include alignment and positioning marks as well as u-shaped silicon channels ending with a 400µm inverse taper length and a with a tip dimensions of 110nm width and 220nm height. The ITPB was applied on this wafer via a commercial high precision wafer level imprint technology at sub-µm placement accuracy (Fig. 4(b)). Fig. 4(c) shows a close up picture of the overlap area of the ITPB and silicon channel. The imprint material used was an ultra-violate curable adhesive with 1.56 index of refraction. After the imprint process, metal was deposited on the turning and curved mirrors (Evaporation process, both gold and aluminum coating were tested). Then the ITPB waveguides were encapsulated with a cladding material at 1.45 index of refraction. Fig. 4(d) is a SEM image which shows the ITPB's waveguide end, the turning mirror and the curved mirror. The calculated optical coupling efficiencies from the silicon channel to the imprinted waveguide for the physical dimensions of silicon channel and the imprinted waveguide listed above, were 98% for the TE mode and 88% for the TM mode (Eigenmodes expansion (EME) simulation method was used).

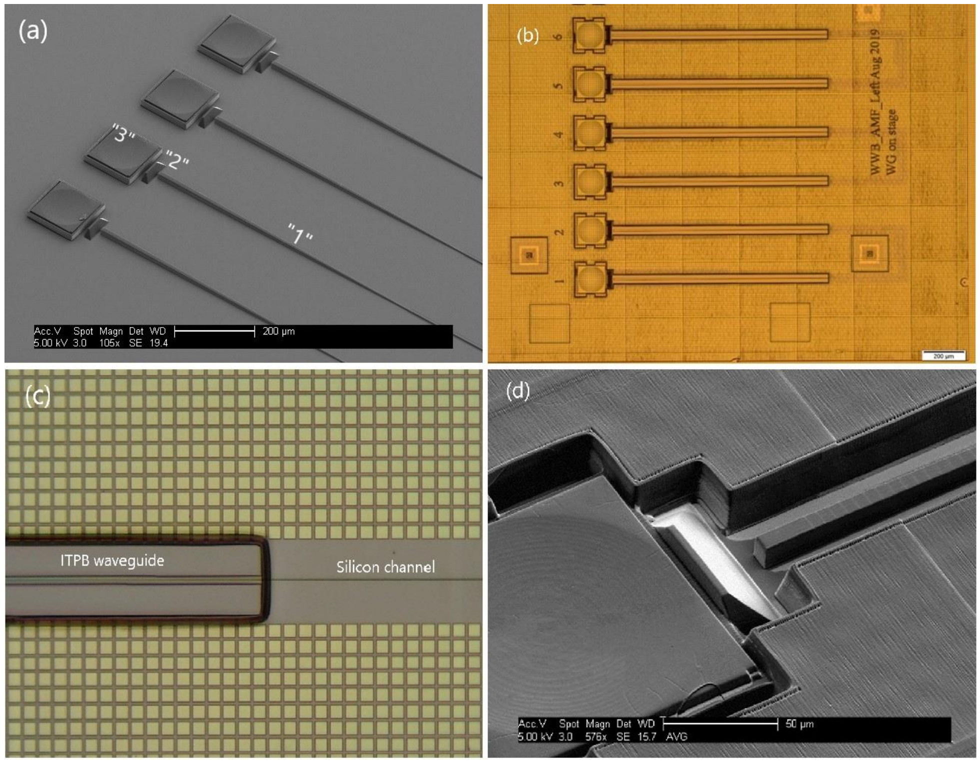

Figure 4: (a) An imprint of inverse taper wide band Photonic-bump (ITPB) structure including waveguide "1", turning mirror "2" and curved mirror "3" (bar is 200µm). (b) A wafer level imprint of ITPB element on AMF silicon photonics test wafer (bar is 200µm). (c) A close up picture, showing the overlap area of the ITPB waveguide and silicon channel. (d) SEM image of an ITPB channel, showing the positive tapered end facet, the turning mirror and the curved mirror (bar is 50µm).

2.3. Tolerance and insertion loss simulations

We used beam tracing and Fast Fourier Transform Beam Propagation Method [FFT-BPM] simulations to evaluate the susceptibility of the system to spatial misplacement of the Photonic-plug in XYZ, to rotations and to tilts. While the simulation, shown in Fig. 1(b) in section 1 above, predicts expanded tolerances values (all stationary parameters at nominal values), the tilts allowed in the system are narrowed as shown in Fig. 5(a) which presents the tilt dependency (with XYZ misplacement at nominal values). This increased sensitivity to tilts is due to the expanded parallel beam propagation between the two curved mirrors. This prediction means that the leveling of the Photonic-plug device is the critical parameter to control in the assembly process. However, given the relatively large size of the Photonic-plug which can reach a few mm's (e.g. 3mmx2mm for a Photonic-plug with 8 fibers), sub-mRad values can be easily achieved by simple incorporation of pedestal structures in the Photonic-plug device. Leveling can be even more accurate as the fiber count rises and the Photonic-plug size is consequently increased. Besides placement and tilt accuracy, the following parameters of reflectivity, shape accuracy and surface quality of the mirrors are other sources for loss contribution. The Photonic-plug's gold coated mirror have a measured roughness of ~10nm RMS and the reflectivity imperfection should contribute 0.13 dB to the loss for every mirror. Therefore, three mirrors will contribute a loss of 0.4dB at perfect placement and tilt accuracies. This penalty is the baseline for the Photonic-plug performance. While this loss is not negligible, it is comparable to the added loss which is often measured post epoxy cure after active alignment assembly at fiber block direct coupling. The following assumptions were considered to assess the typical performance of the Photonic-plug, mirrors focal distance inaccuracy at 5%, placement tilt value at 0.5mRad and the baseline mirrors reflectivity penalty (0.4dB). These loss contributions were added to the simulation and the results of XY and Z tolerances are presented in Fig. 5(b) and Fig. 5(c) respectively.

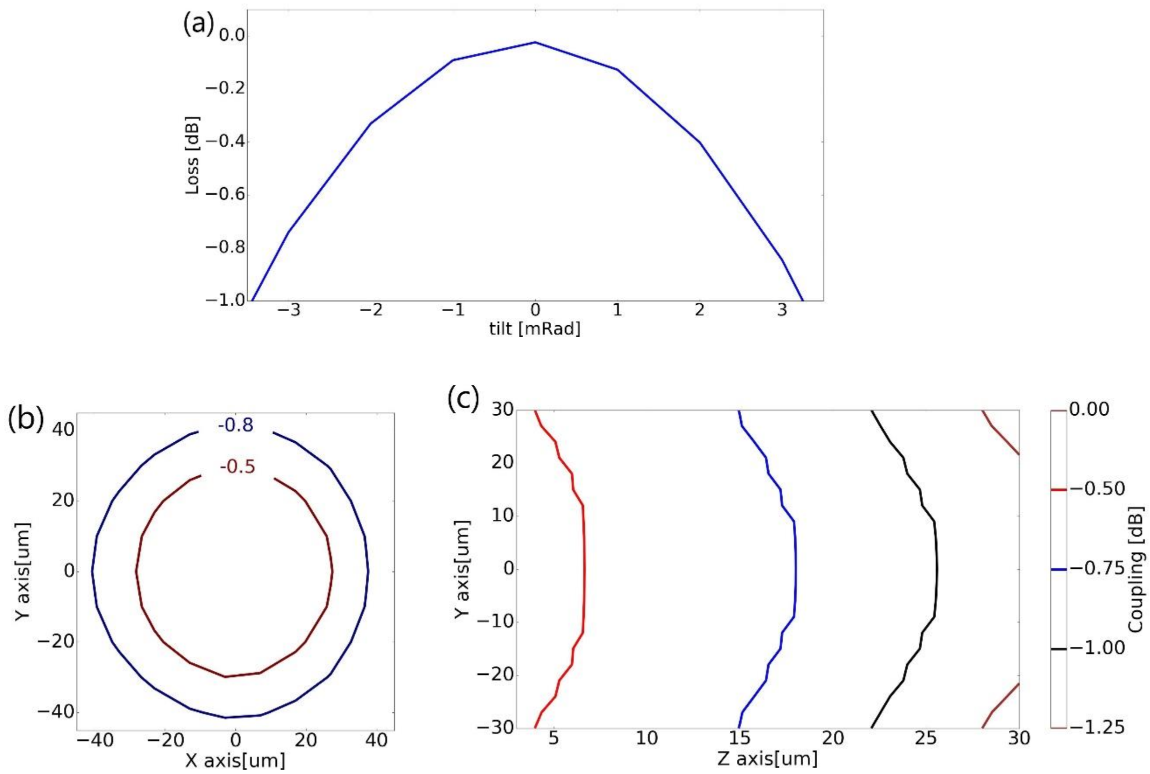

Figure 5: (a) Simulated insertion loss versus tilt dependency of a Photonic-plug device's placement on a silicon photonic chip considering XYZ placements at nominal values. (b) A simulated XY tolerance plot of the Photonic-plug shows 0.5dB and 0.8dB contours. (c) A simulated YZ tolerance plot (Z axis is perpendicular to silicon photonic chip surface). Both simulations in (b) and (c) consider typical performance of the Photonic-plug including of 5% focal distance inaccuracy, 0.5mRad tilt and 0.4dB accumulated loss due to mirrors reflectivity imperfection.

3. EXPERIMENTAL RESULTS

3.1. Photonic-plug tolerance measurements

Spatial tolerance measurements of the Photonic-plug have been performed on a LETI-CEA silicon photonic die and compared to direct fiber coupling to the same chip for the evaluation of assembly tolerances and coupling efficiency parameters. The silicon photonic die included a simple circuit of u-shaped channel with grating couplers for light coupling in and out of the channel. Spatial horizontal and vertical scans of the Photonic-plug have been performed on wafer level top of the test die by an XYZ motorized stage while measuring the coupled light intensity (XY are horizontal plan and Z is vertical to the test die surface). Similar scans on the same test die were obtained by a fiber array block as a reference for Photonic-plug's performance characterization. Fig. 6(a) and Fig. 6(b) show the XY tolerance results for the fiber array block and for the Photonic-plug device (respectively) at 70µmx70µm scan area and a normalized intensity to a single channel of the test die. The -1dB contour data of these scans are shown in Fig. 6(c) and Fig. 6(d), present X and Y tolerance improvement by an order of magnitude and an improvement of total applicable area by almost two orders of magnitude (88 times the area size compared to direct fiber coupling). Fig. 6(d) shows the measured Z tolerance profile performed as a vertical scan above the test die.

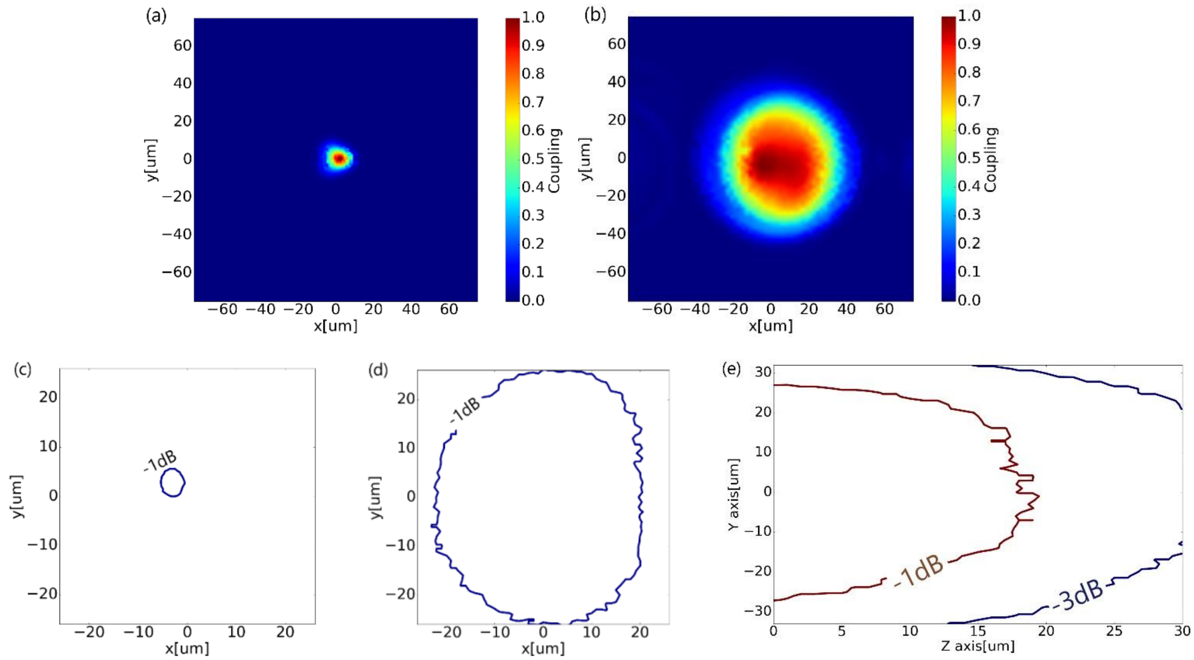

Figure 6: (a) A 70x70µm XY coupling efficiency scan performed by fiber array block directly coupled to the test die input and output channels. (b) A 70x70µm XY coupling efficiency scan performed by the Photonic-plug device on top of the same die. (c) -1dB contour plot of direct fiber coupling shows ±2.5µm XY tolerance. (d) -1dB contour plot of Photonic-plug device coupling shows an XY tolerance larger than ±20µm. (e) A measured Z tolerance obtained via YZ scan above test chip surface.

3.2. Photonic-plug insertion loss measurements

Optical coupling efficiency of the Photonic-plug device has been characterized and compared to direct optical coupling of a fiber array block. The measurements were performed on LETI-CEA grating coupler based die as described in section 2.2 above. The results in Fig. 7 present the measured coupling efficiency of the Photonic-plug (red plot) and of a direct fiber block (blue plot) at lateral positions. The Photonic-plug profile shows a relative 0.5dB insertion loss over a staggering 40µm lateral positioning and 0.6dB inherent loss with reference to direct fiber coupling. These results show one side of the u-shaped channel and do not include grating coupler or channels loss to isolate the Photonic-plug coupling efficiency. Both c-band and o-band grating couplers have been tested and produced similar results. The inherent loss of the Photonic-plug can be further reduced by improving the mirror reflectance by means such as replacing the metal coating by dielectric mirrors layers.

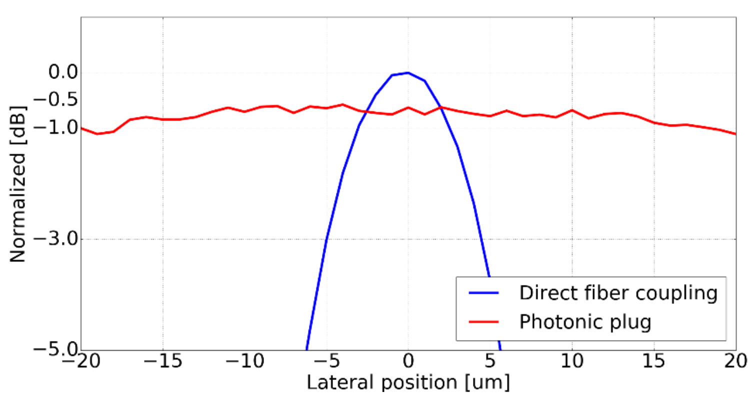

Figure 7: Optical coupling efficiency of the Photonic-plug device (red line) compared to direct coupling via fiber block (blue line) presents 0.5dB insertion loss over range of 40µm in lateral positioning and ~0.6dB as the Photonic-plug's inherent loss.

3.3. Wideband surface coupling measurements

The beam expansion functionality of the ITPB waveguide was evaluated after its imprint on a wideband silicon photonic test chip (AMF, Singapore) as described in section 2.2. A SEM image of the ITPB waveguide is shown in Fig. 8(a) and presents the waveguide's three dimensional structure and its facet's cross section. After beam extraction from the silicon channel of the test chip, the ITPB waveguide expands the beam from 5μmX2μm to 11μmX11μm size. The beam shape at the output of the ITPB waveguide was measured by scanning an SMF parallel to the fact at 1550nm wavelength and presents an expanded beam shape for coupling with SMF fiber (Fig. 8(b)).

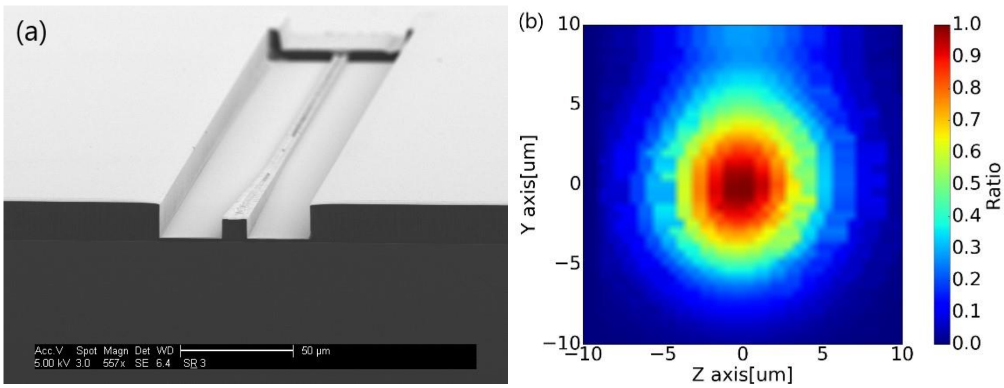

Figure 8: (a) A SEM images presenting three-dimensional ITPB waveguide. This structure was used to characterize the SSC functionality of the ITPB waveguide. (b) A measurement of the spot shape at the ITPB waveguide facet obtained by scanning of SMF at the output of the ITPB facet.

The coupling efficiency from the ITPB's waveguide to the u-shaped photonic test circuit was measured and characterized. For this measurement, a fiber block was positioned in front of the ITPB waveguide's facet. Light was injected through one fiber to the imprinted ITPB waveguide at both polarizations at c-band and then collected by the neighboring fiber to a detector. In its travel, the light passed through two SSC's and two inverted taper silicon sections of the u-shaped silicon channel. The loss of a one way trip (from fiber to silicon), extracted from this measurement, is shown in Fig. 9 and presents a loss of about 1dB for both TE and TM polarizations. Similar measurement setup was constructed to collect light from the silicon photonic chip's surface by utilizing the turning mirror of the ITPB element. These measurements demonstrate the unique performance of the ITPB element to achromatically equalize the TE and TM polarization modes for full wideband surface coupling geometry. These achievements highlight the advantage of the Photonic-plug technology to transfer edge coupling to a convenient surface coupling geometry. These measurements will be followed by a demonstration of a wideband surface coupling at both c-band and o-band with expanded fiber assembly tolerances using the self-alignment optics scheme of the Photonic-plug technology.

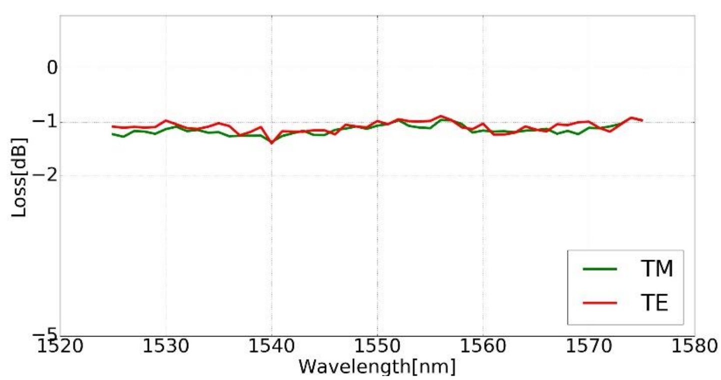

Figure 9: A spectral measurement of ITPB waveguide coupling efficiency presenting a coupling loss of about -1dB at c-band for both TE and TM polarization.

4. CONCLUSIONS

The Photonic-plug technology presents a variety of breakthroughs to enable a solution for scalable packaging of silicon photonics to single mode fiber arrays. Its successful combination of large assembly tolerances and surface coupling geometry, including at wideband, to silicon photonic chips, demonstrates the potential for using high throughput flip-chip assembly tools with passive alignment protocols for silicon photonics packaging. The enablement of surface coupling for all silicon photonic chip kinds further contributes to the reliability of silicon photonics manufacturing as it permits efficient wafer level testing. The ability of the Photonic-plug technology to involve wafer level processes for the accurate placement of optical elements indicates its advantage of using standard wafer manufacturing and benefiting from high volume and reliable fabrication capabilities. The achievements presented in this work mark the potential of the Photonic-plug technology to scale up the silicon photonics industry to further use cases including large port count connectivity which is inevitable in the co-packaged optics roadmap [27], pluggable connectivity and chip-to-chip platforms.

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